DECK 03 · WEEK 2 · MORNING 1

Sampling, Hypothesis Testing & Regression

From a sample to a population — the inferential toolkit underlying every empirical claim in finance

CFA Level I · Vol 1 Learning Modules 7–11

Roadmap — Sampling, Hypothesis Testing & Simple Linear Regression

6 sections · ~2 hours · Built bottom-up from CFA readings

01	**Sampling distributions & CLT**	Probability vs non-probability sampling · CLT · standard error · bootstrap & jackknife
02	**Confidence intervals**	Reliability factors · t vs z · Sendar Fund 95% CI · sample-size levers
03	**Hypothesis-testing framework**	Six-step process · Type I / Type II · power · p-values · two-tailed t-test case study
04	**Common tests (t, z, chi-sq, F)**	Cheat sheet · single variance · variance ratio · chi-sq test of independence (ETFs)
05	**Simple linear regression**	OLS · four assumptions · ANOVA · R^2 · coefficient tests · CAPM-lite worked example · PIs
06	**Big data & ML wrap**	Three Vs · supervised / unsupervised / deep learning · overfitting · train-validate-test

Learning objectives

By the end of this deck you should be able to:

1. Choose between simple random, stratified, cluster and judgmental sampling, and compute the standard error of a sample mean using the central limit theorem.

2. Build a confidence interval for a population mean using the correct reliability factor, and describe how sample size and α change its width.

3. Run the six-step hypothesis-testing process, distinguishing Type I from Type II errors and p-values from critical values for one- and two-tailed tests.

4. Apply the right test statistic — t, z, chi-square or F — to a question about means, variances, correlations or categorical independence, and know its degrees of freedom.

5. Estimate a simple linear regression, test whether its slope is statistically different from zero, and interpret R^2, the standard error of estimate and prediction intervals — while recognising the core ML concepts that extend OLS to big-data settings.

SECTION 01

Sampling distributions & CLT

Probability sampling is random-selection-based and unbiased; non-probability sampling is faster but risks a non-representative sample

All else equal, probability methods yield more accuracy; non-probability methods trade precision for speed and cost.

Method	Type	How it works	Best when
Simple random	Probability	Each element has equal selection probability	Homogeneous population
Systematic	Probability	Select every kth element from an ordered list	Cannot fully enumerate population
Stratified random	Probability	Split into strata, sample within each proportional to size	Indexing, known risk factors
Cluster (one/two stage)	Probability	Split into clusters, pick clusters, then sample all or a sub-sample	Broad geographic surveys
Convenience	Non-probability	Pick whatever data is easy/cheap to access	Pilot studies, cost

STRATIFIED > SIMPLE RANDOM IN FIXED INCOME

Bond index replication divides the universe by duration, sector, credit quality and coupon, then samples within each stratum proportional to market weight — captures the index's risk exposures with far fewer issues than full replication.

Source: CFA Institute, Curriculum Vol 1, Learning Module 7, pp. 196–204.

The Central Limit Theorem says that for any finite-variance population, the distribution of the sample mean approaches Normal(μ, σ^2/n) as n grows — typically $n \geq 30$ suffices

The CLT is the bedrock of inference — it lets us apply normal-distribution tooling to a sample mean even when the underlying returns are far from normal.

$$\bar{X} \sim N(\mu, \sigma^2/n) \quad \text{as} \quad n \to \infty$$

where: μ = population mean; σ^2 = population variance; n = sample size; $\bar{X}$ = sample mean. Convergence is in distribution.

SHAPE

Regardless of whether the population is skewed, bimodal or heavy-tailed, the sample-mean distribution converges to a bell curve.

CENTER

The sample-mean distribution is centered on the population mean — the sample mean is an UNBIASED estimator of μ.

SPREAD

Variance shrinks by 1/n. Standard deviation shrinks by $1/\sqrt{n}$. This is the "standard error" of the mean.

Source: CFA Institute, Curriculum Vol 1, Learning Module 7, pp. 204–212.

The standard error of the sample mean shrinks like 1/√n — quadrupling the sample size only halves the error

Precision improves with the SQUARE ROOT of n — diminishing returns kick in quickly. In CFA Example 2 the mean absolute error barely moves past n = 200.

$$\sigma_{\bar{X}} = \sigma / \sqrt{n} \quad \text{(when } \sigma \text{ known)} \qquad s_{\bar{X}} = s / \sqrt{n} \quad \text{(when } \sigma \text{ unknown)}$$

where: $s^2 = \Sigma(X_i - \bar{X})^2 / (n-1)$, the sample variance; denominator n−1 gives an unbiased estimator of σ^2.

Sample size n	Random-sample MAE	Stratified-sample MAE	Ratio to n=5 (random)
5	0.297%	0.294%	1.00×
20	0.163%	0.152%	0.55×
100	0.063%	0.071%	0.21×
500	0.019%	0.025%	0.06×
1,000	0.009%	0.008%	0.03×

Source: CFA Institute, Curriculum Vol 1, Learning Module 7, Example 2, pp. 201–204.

Bootstrap and jackknife are computer-intensive resampling methods used when no analytical formula for a statistic's standard error exists

Both methods estimate a sampling distribution FROM the data itself — no assumption on the underlying population distribution required.

BOOTSTRAP

Repeatedly draw B samples of size n WITH replacement from the observed sample. Compute the statistic on each draw. The empirical distribution of those B values approximates the sampling distribution.

Used for: standard error of a Sharpe ratio, drawdown metric, or any non-closed-form estimator.

Strength: works for complex estimators and small samples; no normality assumption.

JACKKNIFE

Leave one observation out at a time, recompute the statistic on the remaining n − 1 observations. Exactly n repetitions.

Used for: bias reduction, and quick standard-error estimation of an estimator.

Strength: deterministic (no randomness in resampling) and cheap to run when n is moderate.

Source: CFA Institute, Curriculum Vol 1, Learning Module 7, pp. 212–216.

SECTION 02

Confidence intervals

A 100(1−α)% confidence interval for μ is the point estimate plus and minus a reliability factor times the standard error — t when σ is unknown, z when σ is known or n is large

In finance σ is almost never known, so the t-distribution is the default. As df → ∞, t → z.

$$CI = \bar{X} \pm t_{\alpha/2,\, n-1} \cdot s / \sqrt{n} \qquad (\, z_{\alpha/2} \text{ when } \sigma \text{ known or } n \text{ large} \,)$$

where: $\bar{X}$ = sample mean; s = sample standard deviation; n = sample size. Reliability factor depends on α and df.

Confidence level	α	$z_{\alpha/2}$	$t_{\alpha/2,\, 10}$	$t_{\alpha/2,\, 30}$	$t_{\alpha/2,\, 120}$
90%	0.10	1.645	1.812	1.697	1.658
95%	0.05	1.960	2.228	2.042	1.980

INTERPRETATION

A 95% CI does NOT mean "probability 0.95 that μ lies in this interval". μ is a fixed number. It means: if we repeatedly drew samples and computed CIs, 95% of those intervals would contain μ.

Source: CFA Institute, Curriculum Vol 1, Learning Module 7, pp. 209–212.

Widening the CI by narrowing α, lengthening σ, or shrinking n is mechanical — what changes interpretation is whether the hypothesized value falls inside or outside the bounds

Building a CI and running a two-tailed test at the same α are two views of the same calculation — if the hypothesized value is outside the CI, the test rejects.

SENDAR EQUITY FUND — 95% CI FOR MEAN MONTHLY RETURN

Sendar Equity Fund, 24 monthly returns: $\bar{X}$ = 1.50%, s = 3.60%. Pricing model implies expected return of 1.10%. Build a 95% CI and check whether 1.10% is inside.

1. Degrees of freedom: n − 1 = 24 − 1 = 23
2. Reliability factor (95%, df=23): $t^*_{0.025,23}$ = 2.069
3. Standard error: s / √n = 3.60 / √24 = 0.7348
4. Margin: t* × SE = 2.069 × 0.7348 = 1.5203
5. 95% CI: 1.50 ± 1.52 → [−0.02% , +3.02%]
6. Since 1.10% lies INSIDE the CI, we cannot distinguish the fund's mean from the pricing-model expectation at 5% significance.

95% CI = [−0.02% , +3.02%] → 1.10% is inside → no evidence of outperformance.

Source: CFA Institute, Curriculum Vol 1, Learning Module 8, Exhibit 5, pp. 224–226.

SECTION 03

Hypothesis-testing framework

Every hypothesis test follows the same six steps — stating hypotheses, choosing a statistic, setting α, writing the decision rule, computing, and deciding

Null H_0 and alternative H_a must be MUTUALLY EXCLUSIVE and COLLECTIVELY EXHAUSTIVE. Typically the "suspected" or "hoped-for" condition is stated as H_a.

1. **State the hypotheses**
   Write H_0 and H_a in terms of the population parameter. Two-tailed (=) or one-tailed (≥, ≤).

2. **Identify the test statistic**
   Pick based on what you're testing — t for mean, χ^2 for variance, F for variance ratio, etc.

3. **Specify the significance level**
   α = probability of Type I error. Standard choices: 10%, 5%, 1%. Lower α = more stringent rejection.

4. **State the decision rule**
   Find critical value(s) for this α and df. Reject H_0 if calculated statistic is more extreme.

5. **Calculate the test statistic**
   Plug the data into the chosen formula. Report the value AND the corresponding p-value.

6. **Make a decision**
   Reject H_0 if stat > critical value (equivalently, $p < \alpha$). Otherwise, fail to reject H_0.

Source: CFA Institute, Curriculum Vol 1, Learning Module 8, Exhibit 1, pp. 219–224.

Type I error rejects a true null (false positive, probability α) and Type II error fails to reject a false null (false negative, probability β) — the two are traded off against each other

Lowering α raises β — and vice versa. The only way to reduce both simultaneously is to raise the sample size n.

Decision	H_0 is true	H_0 is false
Fail to reject H_0	Correct decision (prob 1 − α)	TYPE II error (prob β)
Reject H_0	TYPE I error (prob α)	Correct decision (prob 1 − β = POWER)

TYPE I vs TYPE II

Which error is worse depends on the application. In drug approval, Type I (approving an ineffective drug) is worse. In credit screening, Type II (failing to flag a risky borrower) can be. Consequences drive α.

POWER

Power = P(reject H_0 | H_0 false) = 1 − β. Rises with: larger effect size, larger n, lower σ, higher α. Underpowered studies waste data: they cannot detect real effects.

Source: CFA Institute, Curriculum Vol 1, Learning Module 8, Exhibit 3–4, pp. 222–224.

The p-value is the smallest α at which the null can be rejected — reject when p ≤ α

Two-tailed p-value = twice the one-tail probability beyond |calculated statistic|. Reporting the p-value is more informative than a binary reject/fail-to-reject.

$$p = P(\,|T| \geq |t_calc| \;\mid\; H_0 \text{ true}\,) \qquad \text{Reject } H_0 \text{ iff } p \leq \alpha$$

where: T = test-statistic random variable; t_calc = the realised statistic from the sample.

TWO-TAILED TEST

H_0: $\theta = \theta_0$ vs H_a: $\theta \neq \theta_0$.
Reject if $|t_calc| > t_\{\alpha/2, df\}$.
$p = 2 \times P(\,T > |t_calc|\,)$.

RIGHT-TAIL

H_0: $\theta \leq \theta_0$ vs H_a: $\theta > \theta_0$.
Reject if $t_calc > t_\{\alpha, df\}$.
$p = P(\,T > t_calc\,)$.

LEFT-TAIL

H_0: $\theta \geq \theta_0$ vs H_a: $\theta < \theta_0$.
Reject if $t_calc < -t_\{\alpha, df\}$.
$p = P(\,T < t_calc\,)$.

Source: CFA Institute, Curriculum Vol 1, Learning Module 8, pp. 223–224.

A two-tailed t-test on Sendar Fund's monthly returns cannot reject the 1.10% benchmark at 5% because the t-statistic of 0.544 falls inside the ±2.069 critical band

Exactly mirrors the 95% CI on the previous section — 1.10% inside the CI means the test fails to reject at 5%.

SIX-STEP t-TEST OF A SINGLE MEAN

Sendar Equity Fund: $\bar{X}$ = 1.50%, s = 3.60%, n = 24. Test at α = 5% whether the true mean monthly return differs from the 1.10% pricing-model benchmark.

1. STEP 1 — Hypotheses. H_0: μ = 1.10% vs H_a: μ ≠ 1.10% (two-tailed)
2. STEP 2 — Test statistic. $t = (\bar{X} - \mu_0) / (s/\sqrt{n})$ with df = n − 1 = 23
3. STEP 3 — Significance. α = 5%, so α/2 = 2.5% per tail
4. STEP 4 — Decision rule. Reject if |t_calc| > $t^*_{0.025,23}$ = 2.069
5. STEP 5 — Compute. t = (1.50 − 1.10) / (3.60/√24) = 0.40 / 0.7348 = 0.544
6. STEP 6 — Decide. |0.544| < 2.069 → fail to reject H_0 (p ≈ 0.59).

Fail to reject H_0. No evidence the fund's mean differs from the 1.10% model expectation.

Source: CFA Institute, Curriculum Vol 1, Learning Module 8, Exhibit 5, pp. 224–226.

SECTION 04

Common tests (t, z, x^2, F, ρ)

A one-number cheat sheet maps each question about means, variances, correlations or categorical independence to the right test statistic and degrees of freedom

Step 1 of every test is ALWAYS "which row of this table matches my question?" — the rest is mechanical.

What we test	Test statistic	Distribution	Degrees of freedom
Single mean (σ unknown)	$t = (\bar{X} - \mu_0) / (s/\sqrt{n})$	t	$n - 1$
Difference in means ($\sigma_1=\sigma_2$ unknown)	$t = (\bar{X}_1 - \bar{X}_2) / \sqrt{(s^2_p/n_1 + s^2_p/n_2)}$	t	$n_1 + n_2 - 2$
Mean of paired differences	$t = \bar{d} / (s_d/\sqrt{n})$	t	$n - 1$
Single variance	$\chi^2 = (n - 1)s^2 / \sigma_0^2$	χ^2	$n - 1$
Ratio of two variances	$F = s^2_A / s^2_B$ (larger on top)	F	n_A−1, n_B−1
Correlation equals zero	$t = r\sqrt{(n-2)} / \sqrt{(1 - r^2)}$	t	$n - 2$
Independence of two categoricals	$\chi^2 = \Sigma (O_ij - E_ij)^2 / E_ij$	χ^2	$(r-1)(c-1)$

Source: CFA Institute, Curriculum Vol 1, Learning Module 8, Exhibit 2, p. 221; Learning Module 9, pp. 246–248.

Chi-square tests the variance of a single normal population — the test is one-sided with (n−1) degrees of freedom

Unlike t or z, the chi-square distribution is asymmetric — critical values differ between the upper and lower tails.

$$\chi^2 = (n - 1) \cdot s^2 / \sigma_0^2 \qquad df = n - 1$$

where: s^2 = sample variance; σ_0^2 = hypothesized population variance. Assumes underlying population is normal.

SENDAR FUND — IS THE STANDARD DEVIATION LESS THAN 4%?

Same Sendar Fund: 24 months, s = 3.60% (so s^2 = 12.96). Test at α = 5% whether $\sigma^2 < 16$ (i.e. σ < 4%).

1. H_0: $\sigma^2 \geq 16$ vs H_a: $\sigma^2 < 16$ (left-tail, one-sided)
2. χ^2_calc = (24 − 1) × 12.96 / 16 = 23 × 0.810 = 18.63
3. Critical value (5%, df = 23, lower tail): $\chi^2_{0.95,23}$ = 13.09
4. Decision rule: reject H_0 if χ^2_calc < 13.09
5. Decide: 18.63 > 13.09 → fail to reject.

Not enough evidence to conclude that the standard deviation is below 4%.

Source: CFA Institute, Curriculum Vol 1, Learning Module 8, Exhibit 6, pp. 225–228.

An F-test of equal variances uses the ratio of sample variances — always place the larger on top so the statistic exceeds one

Used to check the homoskedasticity assumption in pooled t-tests, and to compare risk across two funds or regimes.

$$F = s^2_A / s^2_B \qquad df = (n_A - 1, n_B - 1)$$

where: A = numerator sample (larger variance). F is right-skewed, supported on [0, ∞). Assumes both populations normal.

DO TWO EQUITY FUNDS HAVE THE SAME RETURN VOLATILITY?

Fund A: 60 monthly returns, s_A = 15.8%. Fund B: 60 monthly returns, s_B = 14.3%. Test equal variances at α = 5% (two-sided).

1. H_0: $\sigma^2_A = \sigma^2_B$ vs H_a: $\sigma^2_A \neq \sigma^2_B$
2. $F_{calc} = s^2_A / s^2_B = 15.8^2 / 14.3^2 = 249.64 / 204.49 = 1.221$
3. Critical value ($\alpha/2$ = 2.5%, df = 59, 59): $F^*_{0.025} \approx 1.67$
4. Decision: 1.221 < 1.67 → fail to reject H_0.

No statistical evidence that the two funds differ in return volatility.

Source: CFA Institute, Curriculum Vol 1, Learning Module 8, pp. 232–236.

A chi-square test of independence asks whether two categorical classifications are related — reject if the calculated statistic exceeds the critical value

Expected frequency under independence: E_ij = (row total × column total) / grand total. Degrees of freedom are (rows − 1) × (cols − 1).

Investment	Small	Medium	Large	Total
Value	50	110	343	503
Growth	42	122	202	366
Blend	56	149	520	725
Total	148	381	1,065	1,594

Observed frequencies of 1,594 ETFs cross-tabulated by size (Small/Medium/Large) and style (Value/Growth/Blend).

CHI-SQUARE TEST OF INDEPENDENCE

H_0: size and style are independent vs H_a: related. $\alpha = 5\%$.

1. Expected E_ij = (row × col) / 1,594. Example (Small, Value): 148 × 503 / 1,594 = 46.7
2. Compute $\Sigma (O_{ij} - E_{ij})^2 / E_{ij}$ across all 9 cells
3. χ^2_calc = 32.08
4. Critical: $\chi^2_{0.05}$, df=4 = 9.488
5. 32.08 > 9.488 → reject H_0

Size and investment style are related (not independent).

Source: CFA Institute, Curriculum Vol 1, Learning Module 9, Exhibits 7–9, pp. 255–261.

SECTION 05

Simple linear regression

Simple linear regression fits a straight line $Y = b_0 + b_1X + \varepsilon$ by minimizing the sum of squared residuals — the least-squares criterion

b_0 = intercept (Y when X = 0). b_1 = slope (change in Y per unit change in X). Residual $e_i = Y_i - \hat{Y}_i$.

$$\hat{b}_1 = \mathrm{Cov}(X, Y) / \mathrm{Var}(X) \qquad \hat{b}_0 = \bar{Y} - \hat{b}_1 \bar{X}$$

$$\text{minimize} \quad \Sigma e_i^2$$

where: Least-squares normal equations produce unique closed-form solutions for $\hat{b}_0$ and $\hat{b}_1$ given a sample of paired (X, Y).

DEPENDENT vs INDEPENDENT

Y (dependent / explained) goes on the vertical axis. X (independent / explanatory) on the horizontal. Choose X because theory or plots suggest it moves Y.

RESIDUAL vs ERROR

ERROR ε: unobserved, from the true population line.
RESIDUAL e: observed, from the fitted sample line.
Least squares makes Σ e = 0 by construction.

Source: CFA Institute, Curriculum Vol 1, Learning Module 10, pp. 266–276.

OLS of monthly equity returns on the market index produces a CAPM-lite alpha (intercept) and beta (slope), interpretable as risk-adjusted excess return and market sensitivity

Running this regression over rolling windows is how practitioners estimate changing betas. CAPM-lite ignores the risk-free rate; full CAPM regresses excess returns on excess market returns.

OLS OF STOCK MONTHLY RETURNS ON S&P 500 — 12 OBSERVATIONS

Stock Y (avg 0.90%/mo) regressed on market X (avg 0.54%/mo). $\Sigma(X - \bar{X})^2 = 48.4$; $\Sigma(X - \bar{X})(Y - \bar{Y}) = 54.2$. Compute intercept and slope.

1. Slope (beta): $\hat{b}_1 = \Sigma(X - \bar{X})(Y - \bar{Y}) / \Sigma(X - \bar{X})^2 = 54.2 / 48.4 = 1.12$
2. Intercept (monthly alpha): $\hat{b}_0 = \bar{Y} - \hat{b}_1 \bar{X} = 0.90\% - 1.12 \times 0.54\% = 0.90\% - 0.60\% = 0.30\%$
3. Fitted line: $\hat{Y} = 0.30\% + 1.12 \times X$
4. Interpretation: for every 1% move in the market, the stock moves ~1.12% on average — high-beta name.
5. Annualised alpha ≈ 0.30% × 12 = 3.6% per year before testing for significance.

$\hat{\beta} = 1.12$ · $\hat{\alpha} = 0.30\%$ / month (≈ 3.6% p.a.) — significance still to be tested.

Source: CFA Institute, Curriculum Vol 1, Learning Module 10, §2 Estimation, pp. 268–276.

OLS requires four assumptions — linearity, homoskedasticity, independence and normality — each testable by looking at the residual plot

Coefficient estimates are unbiased when 1 & 3 hold; STANDARD ERRORS (hence t-tests and CIs) rely on all four.

1. **Linearity**
   E[Y|X] is a straight line in X. Residuals plotted against X should show no curvature. If U- or S-shaped, transform X or fit a non-linear form.

2. **Homoskedasticity**
   Constant residual variance across X. A fan-shape in the residual plot signals heteroskedasticity — use robust standard errors or WLS.

3. **Independence**
   Residuals are uncorrelated across observations. In time series, check for autocorrelation (Durbin-Watson or ACF). Common violator: omitted trend.

4. **Normality**
   Residuals are normally distributed. Matters for inference on small samples; in large samples, CLT makes this assumption forgiving.

Source: CFA Institute, Curriculum Vol 1, Learning Module 10, pp. 278–284.

Total variation decomposes into explained (SSR) plus unexplained (SSE) — R^2 is the explained fraction and the F-statistic tests whether it differs from zero

For simple linear regression, R^2 equals the square of the correlation between X and Y. The F-test of the whole model is equivalent to the t-test on the slope.

$$SST = SSR + SSE \qquad R^2 = SSR / SST \qquad F = MSR / MSE = SSR / (SSE / (n - 2))$$

where: $SST = \Sigma(Y - \bar{Y})^2$; $SSR = \Sigma(\hat{Y} - \bar{Y})^2$; $SSE = \Sigma(Y - \hat{Y})^2$. $MSR = SSR/1$; $MSE = SSE/(n-2)$. F df = (1, n−2).

Source	Sum of squares	df	Mean square	F
Regression (SSR)	191.625	1	191.625 (MSR)	16.35
Error (SSE)	46.875	4	11.719 (MSE)	
Total (SST)	238.500	5		
R^2 = SSR/SST = 0.8035	**s_e = √MSE = 3.423**		**Critical $F_{0.05,1,4}$ = 7.71**	**Reject H_0**

Source: CFA Institute, Curriculum Vol 1, Learning Module 10, §4 ANOVA, pp. 290–298.

The slope's t-statistic tests whether X explains Y — reject H_0: $b_1 = 0$ when $|t|$ exceeds the critical value with n − 2 degrees of freedom

The t-test on a single slope is the workhorse regression test. The same machinery tests whether the slope equals any hypothesised value, not just zero.

$$t = (\hat{b}_1 - b_{1,hyp}) / s_{\hat{b}_1} \qquad df = n - 2$$

where: $s_{\hat{b}_1}$ = standard error of the slope = $s_e / \sqrt{\Sigma(X - \bar{X})^2}$. Confidence interval for b_1: $\hat{b}_1 \pm t^* \cdot s_{\hat{b}_1}$.

ROA REGRESSED ON CAPEX — 6 COMPANIES (CFA LM10)

Estimated slope $\hat{b}_1 = 1.25$. Standard error of slope $s_{\hat{b}_1} = 0.309$. Sample size n = 6. Test H_0: $b_1 = 0$ at $\alpha = 5\%$.

1. df = n − 2 = 4
2. Critical value: $t^*_{0.025, 4} = 2.776$
3. $t_{calc} = (1.25 - 0) / 0.309 = 4.044$
4. $|4.044| > 2.776 \rightarrow$ reject H_0

CAPEX has a statistically significant positive effect on ROA at 5%.

Source: CFA Institute, Curriculum Vol 1, Learning Module 10, §5 Hypothesis Tests, pp. 285–293.

A prediction interval around ŷ widens the further X_f is from x̄ — forecast uncertainty is lowest at the sample's centre of gravity

The PI accounts for BOTH the uncertainty in the fitted line AND the irreducible residual noise — it is always wider than a confidence interval for the mean at X_f.

$$s_f = s_e \cdot \sqrt{[\,1 + 1/n + (X_f - \bar{X})^2 / \Sigma(X - \bar{X})^2\,]} \qquad PI = \hat{Y}_f \pm t^* \cdot s_f$$

where: s_e = standard error of estimate; s_f = forecast standard error; X_f = value of X for which Y is being predicted.

PREDICT ROA AT CAPEX = 6% — CONTINUING ROA/CAPEX EXAMPLE

From earlier: $\hat{Y} = 4.875 + 1.25 \cdot X$; $s_e = 3.423$; $\bar{X} = 6.1$; $\Sigma(X - \bar{X})^2 = 122.64$; $n = 6$. Forecast ROA at $X_f = 6\%$ with a 95% PI.

1. Point forecast: $\hat{Y}_f = 4.875 + 1.25 \times 6 = 12.375\%$
2. Forecast SE: $s_f = 3.423 \cdot \sqrt{1 + 1/6 + (6 - 6.1)^2 / 122.64} = 3.423 \cdot \sqrt{1.1668} = 3.70$
3. Critical $t^*_{0.025,\,4} = 2.776$
4. 95% PI: $12.375 \pm 2.776 \times 3.70 = 12.375 \pm 10.27 \rightarrow [2.1\%, 22.6\%]$

Point forecast 12.4% ; 95% prediction interval [2.1%, 22.6%]. Wide — small sample, high residual noise.

Source: CFA Institute, Curriculum Vol 1, Learning Module 10, §6 Prediction, pp. 299–305.

SECTION 06

Big data & ML wrap

Big data is defined by the three V's — volume, velocity and variety — with a fourth V, veracity, invoked when data are used for inference

Alternative data (social-media text, satellite imagery, credit-card receipts) is growing far faster than traditional financial data (prices, filings).

VOLUME

Megabytes → Gigabytes → Terabytes → Petabytes. Millions to billions of data points per dataset.

VELOCITY

Real-time or near-real-time ticks, clickstreams, sensor feeds. Batch and periodic updates persist but are no longer the norm.

VARIETY

Structured (SQL), semi-structured (HTML, JSON), unstructured (video, audio, text). Tooling must handle all three.

THE FOURTH V · VERACITY

Alternative data sources carry variable credibility: scraped reviews, unverified social posts, satellite mislabels. Assessing source trustworthiness is now a first-order analytical step, not an afterthought.

Source: CFA Institute, Curriculum Vol 1, Learning Module 11, pp. 330–334.

Machine learning splits into supervised (labelled targets), unsupervised (no labels) and deep (multi-layer neural nets) — the right class depends on whether you have a target variable

"Find the pattern, apply the pattern" — Kissell & Mack's one-line definition of ML. Relevant to investing only when enough training data exists.

Class	Training data	Example finance use
Supervised learning	Labelled inputs AND outputs	Predict next-day return direction; forecast credit losses
Unsupervised learning	Inputs only, no labels	Cluster stocks into peer groups; detect outlier trade patterns
Deep learning (neural nets)	Either (hidden layers find features)	NLP on earnings calls; image recognition on satellite data

SLR IS SUPERVISED LEARNING

Our simple linear regression is the simplest possible supervised-learning algorithm — the (X, Y) pairs are labelled and the model learns a mapping. ML generalises this to many non-linear functional forms, but the bias-variance trade-off and the train/validate/test discipline are universal.

Source: CFA Institute, Curriculum Vol 1, Learning Module 11, pp. 335–338.

Overfitting is the cardinal ML sin — a model that fits training data perfectly and fails on the test set; cure is regularization, more data or a simpler model

Always split the data into three: train to fit, validate to tune hyperparameters, test to measure true out-of-sample performance. Never touch the test set during modelling.

OVERFITTING · TOO COMPLEX

Model treats random noise as real signal. Near-perfect fit in training; terrible out-of-sample. Common with too many parameters, too few observations, no regularization.

UNDERFITTING · TOO SIMPLE

Model treats real signal as noise. Poor fit in training AND in test. Common when functional form is too restrictive (e.g. linear fit to a clearly non-linear pattern).

Split	Role	Typical share
Training set	Fit model parameters	60%
Validation set	Tune hyperparameters; compare model families	20%
Test set	Report honest out-of-sample performance	20%

Source: CFA Institute, Curriculum Vol 1, Learning Module 11, pp. 335–338.

KEY TAKEAWAYS

Deck 3 — Sampling, Hypothesis Testing & Simple Linear Regression

- Probability sampling (simple, systematic, stratified, cluster) yields unbiased estimates; non-probability sampling (convenience, judgmental) trades accuracy for speed.
- CI for $\mu = \bar{X} \pm t^* \cdot s/\sqrt{n}$; use t when σ unknown (the default), z only when σ known or n is very large.
- Type I (α) and Type II (β) errors trade off against each other — power = 1 − β rises with n.
- OLS: $\hat{b}_1 = Cov(X,Y)/Var(X)$, $\hat{b}_0 = \bar{Y} - \hat{b}_1\bar{X}$. $R^2 = SSR/SST$; slope test $t = \hat{b}_1/s_{\{\hat{b}_1\}}$ with df = n−2.
- CLT: $\bar{X} \sim N(\mu, \sigma^2/n)$ for any finite-variance population once n is large — standard error shrinks with $1/\sqrt{n}$.
- Six steps: state H_0/H_a; pick statistic; set α; decision rule; compute; decide. p-value ≤ α ⇔reject.
- Pick the right test: t (means), χ^2 (one variance, independence), F (variance ratio), t on r (correlation).
- Big data: 3 V's + veracity. ML splits into supervised / unsupervised / deep learning. Overfitting fixed by regularization, more data, or a train-validate-test split.